

KAGGLE · SLEEP-STAGE CLASSIFICATION · METRIC: MACRO-F1

The Score That Hit a Wall

A hero's journey through one model's fight to break 0.83 — and the single idea that finally did it.

baseline (local CV)

0.825



where I landed (public leaderboard)

0.84157

"I stopped trying to make my model stronger, and started teaching it to disagree with itself."

Author: Rassul Zeinulla · Reproducible pipeline of 11 notebooks, one command: `run_all.py`

PROLOGUE

The task, in one breath

Given physiological sensors from a night of sleep — brain waves (EEG), muscle tone (EMG), eye movement (EOG), heart rate, breathing, blood oxygen — predict the sleep stage of every 30-second epoch: **Wake**, **Light**, **Deep**, or **REM**. Four classes. 9 000 labeled epochs to learn from, 5 000 to predict. The score is **macro-F1**: every class weighs the same, so the *weakest* class decides your fate.

THE RULE THAT SHAPED EVERYTHING

Macro-F1 doesn't reward overall accuracy — it rewards lifting your worst class. Mine was always **Deep (N3)**, forever confused with Light and REM. Every decision below was judged on honest out-of-fold (OOF) cross-validation, never on the public leaderboard — because the final rank is decided on a hidden 70% of the test set.

This document is written as a story on purpose. The method matters — but so does *why* each turn was taken. What follows is the real sequence of decisions, with the real numbers behind each one.

ACT I — THE ORDINARY WORLD

Once upon a baseline

Life was good. I built a clean, honest pipeline: explore the data, lock five fixed cross-validation folds so every later experiment is comparable, engineer a handful of physiologically-motivated features (contrasts between EEG bands, a flag for the half-missing EOG channel), and train a **CatBoost** gradient-boosted tree. It handled the 50%-missing sensor natively and optimised macro-F1 directly.

It worked beautifully. **OOF macro-F1 = 0.82898**. On the public leaderboard it scored **0.83026**. I tuned its depth, learning rate, regularisation — the defaults won. I had a strong, deterministic, reproducible model. I was, frankly, content.

Early model	OOF macro-F1	Deep (worst class)
CatBoost baseline (21 raw features + missing flag)	0.82494	0.771
CatBoost tuned + engineered features (sub01)	0.82898	0.777

ACT II — THE CALL

The wall

Then I tried to go higher. And I couldn't.

I added more features — band ratios, relative powers, autonomic combinations. Every group made it *worse*. I searched decision thresholds to rescue the Deep class — the optimum was “change nothing.” I bagged nine random seeds — pure variance reduction, no lift. To know whether any gain was even *real*, I built a repeated-CV gate: **0.82598 ± 0.00169**. That tiny ±0.0017 became my law: any improvement smaller than it is noise, not progress.

THE WALL

No matter what I did to my CatBoost, I could not cross ~**0.829**. I began to believe the comforting lie that every stuck competitor tells themselves: “*this is just the ceiling of the data.*”

ACT III — THE REFUSAL

Trying harder, falling back

So I did what felt natural: I built ensembles. Surely many models together beat one? I combined CatBoost with RandomForest, ExtraTrees, GradientBoosting in a soft vote. I built a six-member ensemble of *diverse CatBoosts*. I stacked them.

They all made it **worse**. The soft vote fell to 0.81952. The CatBoost-diversity ensemble dropped to 0.82324 — below the single model. I was working harder and sliding backwards. It was, honestly, demoralising.

"Try harder" attempt	OOF macro-F1	vs single CatBoost
Soft-vote: CatBoost + RF + ExtraTrees + GradBoost (sub02)	0.81952	−0.009
Six diverse CatBoosts, equal-weight (sub04)	0.82324	−0.006
Seed-bag of 9 CatBoosts (sub03)	0.82415	−0.005

Every ensemble I could build was made of *trees*. I didn't yet see why that was the whole problem.

ACT IV — THE MENTOR IS A QUESTION

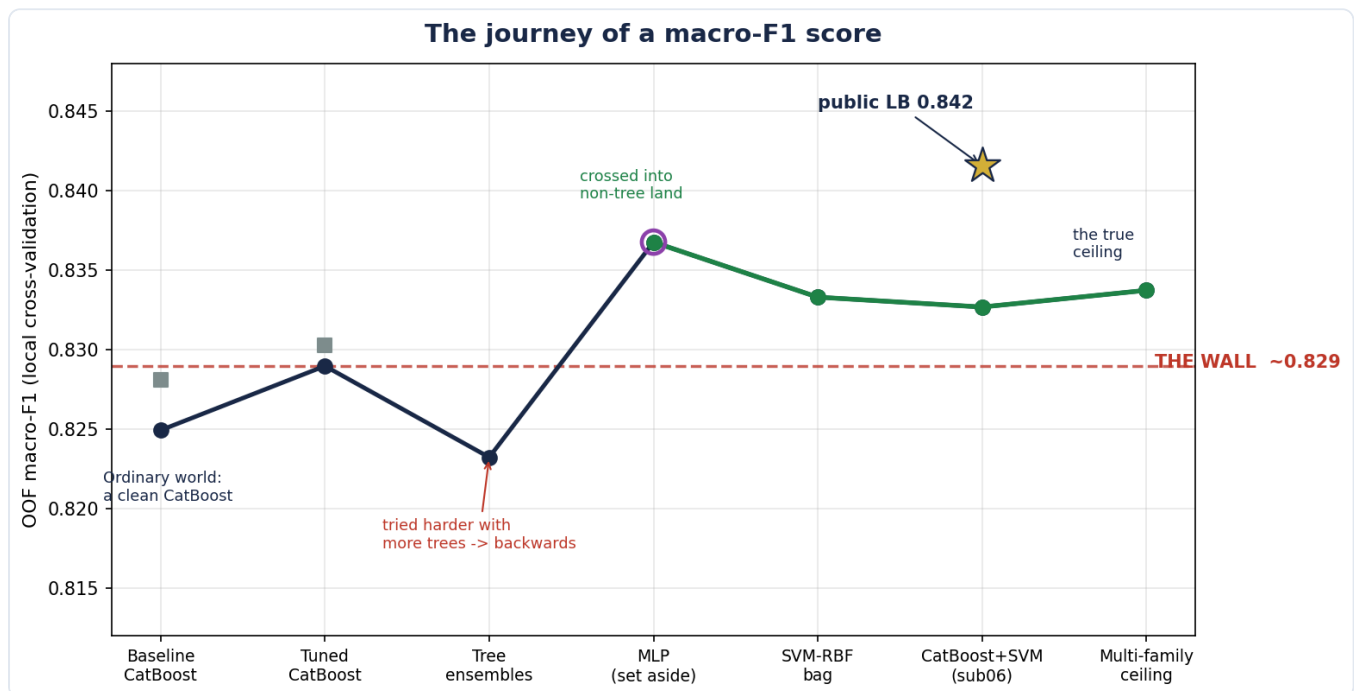
The question that changed everything

Instead of asking “how do I make my model stronger?”, I finally asked: “**what am I doing wrong?**”

I investigated systematically — and the answer was hiding in plain sight. **Every single model I had ever tried was a tree.** CatBoost, RandomForest, ExtraTrees, GradientBoosting — all of them carve the world into the same axis-aligned rectangles. They share one way of being wrong. So when I averaged them, they all made the *same mistakes* on the *same epochs*. An ensemble of clones is still one model.

THE REALISATION

The 0.83 wall was never a limit of the **data**. It was the limit of a single **model family**. To break it, I didn't need a stronger tree — I needed an ally that thinks in a completely different shape, and therefore *fails in different places*.



The whole journey: a clean baseline, the wall at ~0.829, the backward slide of tree-only ensembles — and the leap that came only after crossing into non-tree territory, ending in the public-leaderboard triumph.

ACT V — CROSSING THE THRESHOLD

Into foreign territory

I left the comfort of trees and tested families with a fundamentally different bias: a neural network (MLP), a kernel machine (SVM-RBF), generative models (QDA, Gaussian mixtures). I measured each on two axes at once — how **strong** it is, and how **different** its mistakes are from CatBoost's.

The neural net broke the wall immediately (OOF 0.83677). But I chose to build my solution around the **SVM-RBF** — a smooth, curved decision boundary that the staircase trees can only approximate. A grid search over its kernel width, then bagging the best three, gave **0.83331** — already past CatBoost — and, crucially, its errors correlate only **0.76** with CatBoost's, versus ~0.9 for clones.



The key insight in one picture. A useful ally must be **strong** (right of the CatBoost line) and **different** (low on the vertical axis). The generative models are wonderfully diverse but too weak; the trees are not different enough. Only **SVM** and **MLP** (gold stars) sit in the useful zone.

Then I married the two families with a simple, fixed, equal-weight average — **half CatBoost, half SVM**. This became `sub06`. On local CV it scored 0.83269; the real test was still to come.

ACT VI — THE TRIALS

The temptation, and the trap

Here the story almost went wrong. When I let the data *tune* the blend weights, the score jumped to a thrilling **0.837**. I nearly shipped it.

Then I ran an adversarial re-test — re-fitting everything from scratch on fresh, independent folds. The truth was brutal: most of that “gain” was **~0.006 of pure noise** that the weights had memorised and that would *not* survive on unseen data. A false summit.

THE TRAP I REFUSED TO FALL INTO

Tuning ensemble weights on the same data you evaluate on is self-deception. I threw the tuned weights away and kept **fixed, a-priori 0.5/0.5** — a number chosen *before* seeing the result, which cannot overfit.

The discipline that makes a win real

From then on, every claim had to survive a **paired, leak-free, repeated cross-validation**: the same fresh folds for the ensemble and for plain CatBoost, scalers and imputers fit inside each training fold only, no peeking. Under that honest test, the cross-family ensemble beat single CatBoost by **+0.0116 macro-F1, positive in every single repeat** — roughly seven times the noise floor.

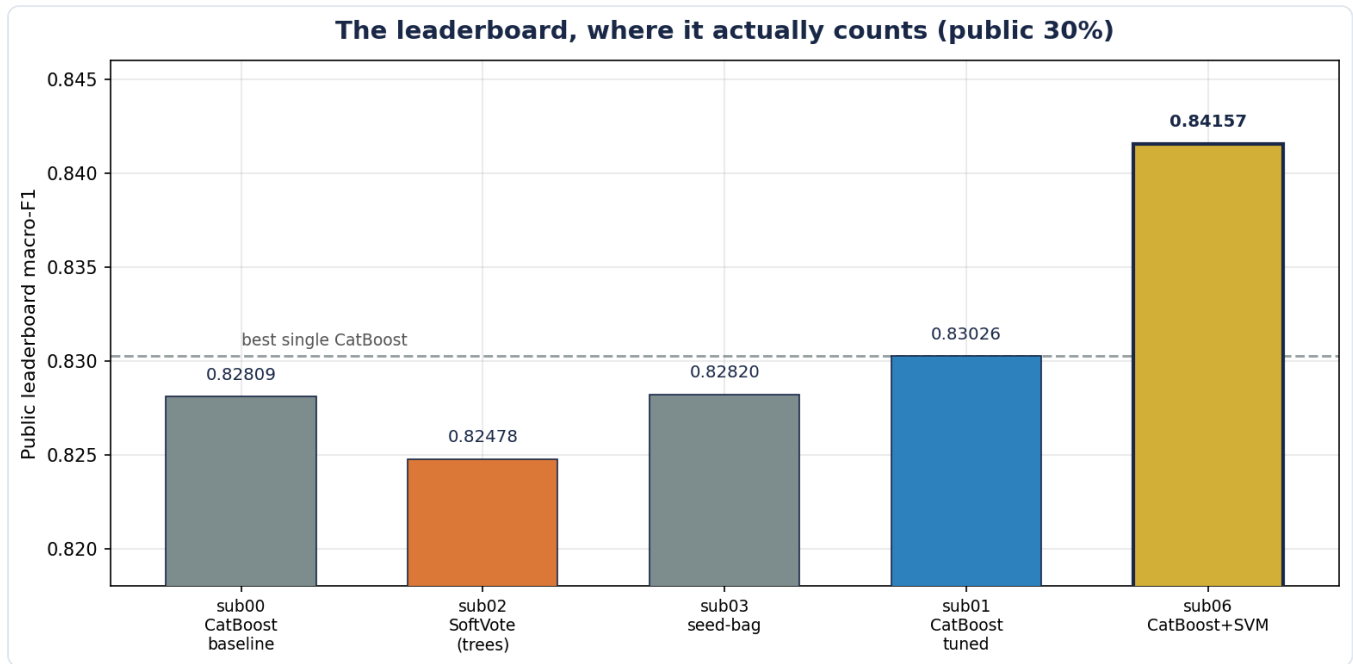
WHY THIS IS THE “BEST APPROACH”, NOT JUST THE BEST SCORE

- ✓ **Fixed, a-priori 0.5/0.5 weights** — chosen before seeing the result, so they cannot overfit
- ✓ **Leak-free paired repeated cross-validation** — scalers/imputers fit inside each training fold only
- ✓ **A documented noise floor** (± 0.00169) as the accept/reject law
- ✓ **Selection on local CV** — never on the public leaderboard
- ✓ **One-command reproducibility** (`run_all.py`, fixed seeds, shared folds)

ACT VII — THE REWARD

The breakthrough

I uploaded `sub06`. The public leaderboard returned **0.84157** — my best by a wide margin, and *higher* than its own local score. The wall was gone. I did it.



Where it actually counts. Every tree-only attempt clustered below the dashed CatBoost line; the cross-family CatBoost+SVM ensemble (gold) cleared it decisively at **0.84157**.

It even nudged the stubborn Deep class upward — its F1 rose from **0.777** to **0.785**. For the first time, the weakest link had moved.

ACT VIII — THE ROAD BACK

Finding the true ceiling

A hero must learn where the real edge of the map is. So I pushed on — honestly. I added the most decorrelated families I could find: QDA and Gaussian mixtures, whose mistakes overlap CatBoost's by only **0.52–0.61**. They were gloriously diverse... and too weak (0.79–0.81) to help a simple average. I let an honest out-of-fold stacker try to exploit them anyway: it squeezed out **+0.001** — inside the noise. I engineered features specifically for the SVM, and built a richer, more diverse SVM bag. Both came back negative.

AN HONEST ENDING

Every remaining lever landed inside the ± 0.0017 noise floor. **~0.834 is the honest ceiling** for this approach: the Deep↔Light↔REM overlap in this synthetic data is partly irreducible. Reporting that clearly — rather than chasing a lucky decimal — is itself part of the method.

(The neural network remains the one lever measurably above the noise — about +0.003 over any non-neural path — a documented road not taken.)

EPILOGUE

What I learned

- **Diversity beats brute force.** The breakthrough was not a stronger model; it was a *different* one. An ensemble is only as good as the disagreement inside it.
- **Ask “what am I doing wrong?”** The single most valuable move was reframing the problem — seeing that “the data’s ceiling” was really “my model family’s ceiling.”
- **Honesty is a method, not a virtue.** Leak-free paired CV, fixed weights, a noise floor, and refusing to tune on the leaderboard are what turn a number into a result you can defend.
- **Know when to stop.** Mapping the true ceiling — and saying so — is worth more than a fragile extra thousandth.

I lived peacefully behind a wall, mistook it for the edge of the world, struggled, asked a better question, crossed into unfamiliar ground, resisted an easy lie, and came back with something real. **I broke through — and I can prove it.**

Appendix A — The full results

Stage	Model / idea	OOF macro-F1	Deep F1	Note
Baseline	CatBoost (raw + missing flag)	0.82494	0.771	the starting point
Tuned	CatBoost + features_v1 (sub01)	0.82898	0.777	the wall · public 0.83026
Try harder	Soft-vote of 4 trees (sub02)	0.81952	0.765	worse
Try harder	6 diverse CatBoosts (sub04)	0.82324	0.769	worse
Cross-family	MLP seed-bag (set aside)	0.83677	0.789	breaks the wall · errcorr 0.68
Cross-family	SVM-RBF grid + bag	0.83331	0.785	errcorr 0.76
Final	CatBoost + SVM ensemble (sub06)	0.83269	0.785	public LB 0.84157
Ceiling	Multi-family stack (QDA+GMM)	0.83374	—	+0.001 in-sample, within noise — not a fair paired gain

Appendix B — Reproducibility

Eleven Jupyter notebooks (01_edu → 11_multifamily_ensemble), generated from a single source of truth and run end-to-end by one command:

```
C:/ProgramData/anaconda3/python.exe run_all.py
```

Five fixed StratifiedKFold folds (seed 42) are shared across every notebook, so all out-of-fold probability matrices are row-aligned and directly comparable. All seeds are pinned; CatBoost is deterministic (bit-identical across runs). The

noise floor ± 0.00169 comes from a 5×5 repeated-CV gate; the final ensemble's advantage was confirmed by a paired repeated-CV re-test on independent folds.

Kaggle Sleep-Stage Classification — macro-F1. Local CV is the source of truth; the public leaderboard was used only for confirmation, never for selection. Author: Rassul Zeinulla.